

PAPER Nuclear-Data Uncertainty Quantification for Tritium Breeding: Which 14 MeV Cross-Section Channels Set the Error Bar on the Breeding Ratio

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Abstract

A tritium breeding ratio (TBR) is only as trustworthy as the cross-sections that produced it, and a design that quotes a TBR without an error bar is quoting a point without a tolerance. We establish the credible nuclear-data uncertainty on the TBR of a compact spherical-tokamak breeder (design point $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $\delta = -0.30$, $B_0 = 8$ T, $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$) by casting the TBR as an integral response functional of the transport problem, deriving its first-order cross-section sensitivities from generalised perturbation theory, and propagating a channel-resolved data uncertainty through the sandwich rule $\sigma_R^2 = \mathbf{S}^\top \mathbf{C} \mathbf{S}$. The load-bearing question is answered channel by channel. Processing evaluated data (ENDF/B-VIII.0) and default-model data (TALYS) identically through NJOY21 and transporting both through the same three-dimensional Monte-Carlo geometry, we measure a model-form spread of 55–66% at 14.06 MeV on the two breeding-relevant light channels — ${}^7\text{Li}(n, n't)$ (66.1%) and ${}^9\text{Be}(n, 2n)$ (55.1%) — against only $\approx 15\%$ on the heavy structural multiplier ${}^{184}\text{W}(n, 2n)$. Because TBR is a near-linear functional of the light reaction rates, these two channels carry $\approx 95\%$ of the model-form variance, and the single largest tritium producer, ${}^6\text{Li}(n, t)$, is an evaluated standard whose covariance is of the few-per-cent class. The cross-library reproduction (ENDF/B-VIII.0 versus FENDL-3.2) agrees to 0.3%, two orders of magnitude below the model spread, proving the code path is not the source of the band — the evaluated data is. The result is a requirement, not a caveat: the light breeding channels must be taken from evaluated covariances, and a TBR reported without propagating them understates the tolerance on the most load-bearing number in the breeder design by more than an order of magnitude. Every headline quantity is a frozen anchor (gate BR-L2-A28) in the de-risking register^[reg].

Keywords: nuclear data; uncertainty quantification; tritium breeding ratio; ENDF/B-VIII.0; TALYS; NJOY21; sensitivity/uncertainty analysis; covariance

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1 Introduction

The tritium breeding ratio is the number on which the entire case for a breeding blanket rests. It is defined as the number of tritons produced per fusion neutron, and for a device whose product is tritium, helium-3 and 14 MeV neutrons rather than net electricity, it is the primary figure of merit: a blanket with $TBR < 1$ consumes its own fuel, and the fleet path to self-sufficiency and export is set entirely by how far above unity the ratio can be driven and held [2, 3]. The compact spherical-tokamak (ST) breeder studied here is a high-power-density, negative-triangularity device operating at $I_p = 9.66$ MA, $Q = 3.076$ and $P_{fus} = 85.04$ MW, in the compact-high-field ST tradition of ST-FNSF, ARC and SPARC [5, 6, 7]. Companion analyses in the same design register establish the net three-dimensional TBR at the operating point, the beryllium and ^6Li levers that walk it above the self-sufficiency floor of 1.05, and the fleet doubling time it implies [30, 32, 33, 34]. Those analyses quote a *central* value. This paper supplies the tolerance around it, and answers the sharper question implied by the title: of the hundreds of reaction channels that a Monte-Carlo transport calculation samples, which ones actually set the error bar on the breeding ratio?

1.1 The problem: a central value is not a claim

A TBR computed to three decimal places is a statement about the evaluated nuclear data as much as about the geometry. Three sources of uncertainty enter a computed integral response: (i) the transport method and its statistical error, (ii) the geometry and material fidelity, and (iii) the evaluated cross-sections and their covariances. The first is controllable to arbitrary precision by spending Monte-Carlo histories; the second is a modelling choice audited elsewhere in the register by CAD-faithful transport [31]; the third is irreducible without new measurement or re-evaluation. It is common in fusion-blanket design to report a central TBR and either omit the nuclear-data component altogether or fold it into a single lumped percentage. Neither is adequate for a load-bearing number: the design needs to know *which* channels dominate, because those are the channels whose evaluations must be trusted, whose covariances must be propagated, and — where the evaluations of the field disagree — which must be targeted by differential or integral measurement.

1.2 Prior art

The machinery for propagating nuclear-data uncertainty to integral responses is mature. Deterministic sensitivity/uncertainty (S/U) analysis via the sandwich rule and adjoint-based generalised perturbation theory is standard in reactor physics and is implemented in tools such as the SCALE/TSUNAMI system [22]. Stochastic propagation — Total Monte Carlo and its fast variants, in which the data itself is sampled from its covariance and the transport re-run — provides an independent cross-check that avoids the first-order linearisation [24, 23]. Evaluated libraries now ship the covariances required to make either approach quantitative: ENDF/B-VIII.0,

JEFF-3.3 and JENDL-5 all carry cross-reaction, cross-energy covariance matrices for the fusion-relevant nuclides [8, 10, 11, 9], and the light-element breeding and standard channels are additionally anchored by the international neutron-data standards evaluation [12] and by the EXFOR compilation of differential measurements [13]. For fusion blankets specifically, S/U studies of the DEMO helium-cooled pebble-bed and helium-cooled lithium-lead concepts have quantified the TBR nuclear-data uncertainty and identified lithium, beryllium, lead and the structural iron chains as the leading contributors [25, 26, 27]. What these studies share — and what we adopt — is the finding that the breeding uncertainty is carried by a small number of light and multiplier channels, not by the bulk of the structural data.

1.3 Contribution

This paper makes the nuclear-data tolerance on the ST breeder’s TBR explicit and channel-resolved, and does so end-to-end from the same processing and transport stack that produced the central value. Specifically, we

1. cast the TBR as an integral response functional of the fixed-source transport problem and derive its first-order cross-section sensitivities from adjoint/generalised perturbation theory (section 2);
2. measure, as a frozen de-risking anchor (gate BR-L2-A28), the model-form spread between evaluated (ENDF/B-VIII.0) and default-model (TALYS) cross-sections for the breeding-relevant channels, processed identically through NJOY21 and transported through the same three-dimensional geometry (sections 3, 4);
3. propagate that spread through the sandwich rule to partition the TBR variance by channel, showing that two light channels carry $\approx 95\%$ of the model-form variance while the voluminous structural data contributes little (section 4);
4. establish that the code path is not the source of the band — the cross-library reproduction agrees to 0.3% — and therefore that the usable TBR error bar must be quoted from evaluated covariances, of the few-per-cent class, not from model defaults, which would put an unusable $\gtrsim 50\%$ band on the ratio (sections 4, 6).

The physical explanation for the channel dependence — the failure of the compound-nucleus statistical model for light nuclei — is developed in section 5, and the GPU-accelerated Monte-Carlo and data-processing campaign that produced the numbers is documented in section 3. The work sits within a broader global sensitivity study of the frozen design point [35] and a verification-first computational programme built on independent-library gate reproduction [36].

2 Governing equations and formal problem

2.1 The breeding ratio as an integral response functional

Neutron transport in the blanket is governed by the time-independent, fixed-source linear Boltzmann equation for the angular flux $\psi(\mathbf{r}, \boldsymbol{\Omega}, E)$,

$$\boldsymbol{\Omega} \cdot \nabla \psi + \Sigma_t(\mathbf{r}, E) \psi = \int_{4\pi} \int_0^\infty \Sigma_s(\mathbf{r}; E' \rightarrow E, \boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}) \psi(\mathbf{r}, \boldsymbol{\Omega}', E') dE' d\boldsymbol{\Omega}' + S(\mathbf{r}, \boldsymbol{\Omega}, E), \quad (1)$$

where Σ_t and Σ_s are the macroscopic total and double-differential scattering cross-sections and S is the fusion source. For a D–T plasma the source is monoenergetic to good approximation at the 14.06 MeV D–T reaction energy, $S(\mathbf{r}, \boldsymbol{\Omega}, E) = \frac{S_0}{4\pi} q(\mathbf{r}) \delta(E - E_0)$ with $E_0 = 14.06$ MeV, q the normalised plasma emissivity and S_0 the total neutron rate. Writing (1) compactly as $\mathcal{L}\psi = S$ with $\mathcal{L}$ the transport–collision operator, the scalar flux is $\phi(\mathbf{r}, E) = \int_{4\pi} \psi d\boldsymbol{\Omega}$.

The tritium breeding ratio is the tritium-production rate per source neutron. Tritons are bred by the exothermic ${}^6\text{Li}(n, t){}^4\text{He}$ reaction and the endothermic ${}^7\text{Li}(n, n't){}^4\text{He}$ reaction, so

$$\text{TBR} = R[\phi] = \frac{1}{S_0} \int_V \int_0^\infty \left[N_6(\mathbf{r}) \sigma_{6t}(E) + N_7(\mathbf{r}) \sigma_{7t}(E) \right] \phi(\mathbf{r}, E) dE dV, \quad (2)$$

where N_6, N_7 are the ${}^6\text{Li}, {}^7\text{Li}$ number densities and $\sigma_{6t} \equiv \sigma_{6\text{Li}(n,t)}$, $\sigma_{7t} \equiv \sigma_{7\text{Li}(n,n't)}$ are the microscopic triton-production cross-sections. The flux itself depends on *every* cross-section in the problem through $\mathcal{L}$; in particular the neutron multiplier ${}^9\text{Be}(n, 2n)$ and ${}^{184}\text{W}(n, 2n)$ raise ϕ and hence raise (2) indirectly, while parasitic captures such as $\text{Fe}(n, \gamma)$ and $\text{W}(n, \gamma)$ depress it. Equation (2) makes explicit that TBR is a bilinear functional of the response cross-sections and the flux, and it is this dual dependence — direct through the response operator, indirect through the flux — that the sensitivity analysis must capture.

2.2 First-order sensitivity from generalised perturbation theory

Let Σ_i denote a generic cross-section (nuclide n , reaction x , energy E), collected into a parameter vector $\mathbf{\Sigma}$. The relative sensitivity of the response to Σ_i is

$$S_{R,i} \equiv \frac{\partial \ln R}{\partial \ln \Sigma_i} = \frac{\Sigma_i}{R} \left[\underbrace{\left. \frac{\partial R}{\partial \Sigma_i} \right|_{\phi}}_{\text{direct}} + \underbrace{\left\langle \frac{\partial R}{\partial \phi}, \frac{\partial \phi}{\partial \Sigma_i} \right\rangle}_{\text{indirect}} \right]. \quad (3)$$

The direct term is non-zero only for the two triton-production reactions appearing in (2). The indirect term requires the perturbed flux $\partial \phi / \partial \Sigma_i$; rather than solve a forward problem for every Σ_i , generalised perturbation theory introduces the adjoint (importance) function $\psi^\dagger$ solving

$$\mathcal{L}^\dagger \psi^\dagger = \Sigma_R, \quad \Sigma_R(\mathbf{r}, E) = N_6 \sigma_{6t}(E) + N_7 \sigma_{7t}(E), \quad (4)$$

with $\mathcal{L}^\dagger$ the adjoint transport operator and the response cross-section Σ_R acting as the adjoint source. Differentiating $\mathcal{L}\psi = S$, taking the inner product with $\psi^\dagger$ and using the adjoint identity $\langle \psi^\dagger, \mathcal{L} \delta \psi \rangle = \langle \mathcal{L}^\dagger \psi^\dagger, \delta \psi \rangle$ gives the indirect sensitivity in a single adjoint solve,

$$S_{R,i}^{\text{indir}} = -\frac{\Sigma_i}{R} \left\langle \psi^\dagger, \frac{\partial \mathcal{L}}{\partial \Sigma_i} \psi \right\rangle = -\frac{\Sigma_i}{R} \left\langle \psi^\dagger, \left(\frac{\partial \Sigma_t}{\partial \Sigma_i} - \frac{\partial \Sigma_s}{\partial \Sigma_i} \right) \psi \right\rangle. \quad (5)$$

The total sensitivity $S_{R,i} = S_{R,i}^{\text{dir}} + S_{R,i}^{\text{indir}}$ is dimensionless and, integrated over energy, defines the channel-integral sensitivity $S_c = \sum_g S_g^c$ for channel $c = (n, x)$. Two consequences follow that drive the whole result. First, for the triton-production channels the direct term dominates and is *positive*: to leading order $S_{6t} \simeq T_6/\text{TBR}$ and $S_{7t} \simeq T_7/\text{TBR}$, the fractional contribution of each lithium isotope to the bred inventory. Second, for the $(n, 2n)$ multipliers the sensitivity is purely indirect but strong, because a single multiplication event feeds *both* lithium channels; for parasitic captures it is small and negative. The near-linearity of (2) in the light reaction rates is what makes the light channels simultaneously the most sensitive and, as we show, the most uncertain.

2.3 The sandwich rule and the covariance structure

Given the sensitivity vector $\mathbf{S} = \{S_{R,i}\}$ and the relative covariance matrix $\mathbf{C}$ of the evaluated cross-sections, the first-order (linear-propagation) variance of the response is the quadratic form

$$\left(\frac{\sigma_R}{R} \right)^2 = \mathbf{S}^\top \mathbf{C} \mathbf{S} = \sum_{i,j} S_{R,i} C_{ij} S_{R,j}. \quad (6)$$

The covariance carries the physics of how the data uncertainties correlate. Evaluated covariances are block structured: a diagonal block $\mathbf{C}^{n,xx}$ for each (nuclide, reaction) pair, an energy-group $g \times g'$ matrix whose off-diagonals encode the strong energy correlation of an evaluation; off-diagonal blocks $\mathbf{C}^{n,xx'}$ for cross-reaction correlations within a nuclide (e.g. elastic–inelastic sum rules); and, in the standards sub-library, cross-*material* correlations induced by shared reference reactions [12]. Writing the sensitivity of channel $c = (n, x)$ as the group vector $\mathbf{S}^c = \{S_g^c\}$, (6) decomposes as

$$\left(\frac{\sigma_R}{R} \right)^2 = \sum_n \sum_{x,x'} \mathbf{S}^{n,x\top} \mathbf{C}^{n,xx'} \mathbf{S}^{n,x'} \equiv \sum_c V_c + \sum_{c \neq c'} \text{Cov}_{cc'}, \quad (7)$$

where the *partial variance* of channel c is $V_c = \mathbf{S}^{c\top} \mathbf{C}^{cc} \mathbf{S}^c$ and the second sum collects cross-channel covariances. The fraction of the TBR variance owned by a channel is $f_c = V_c / \sigma_R^2$. The engineering question “which channels set the error bar” is precisely: for which c is f_c of order unity?

2.4 The model-form spread as a bounding proxy

A complete answer to (6) requires the full evaluated $\mathbf{C}$. A conservative and provenance-honest proxy for the *model-form* component of the uncertainty — the part that reflects our ignorance of the true cross-section shape, as opposed to the reported evaluation uncertainty — is the discrepancy between an experiment-anchored evaluation and a purely default-model calculation of the same channel. Define the fractional model spread

$$d_c \equiv \frac{\Sigma_c^{\text{model}} - \Sigma_c^{\text{eval}}}{\Sigma_c^{\text{eval}}}, \quad \Sigma_c^{\text{model}} = \text{TALYS default}, \quad \Sigma_c^{\text{eval}} = \text{ENDF/B-VIII.0}, \quad (8)$$

both evaluated at the source energy and both processed identically to macroscopic form. Treating $|d_c|$ as a 1σ , fully energy-correlated proxy for channel c and neglecting cross-channel correlation gives the uncorrelated model-form band

$$\left(\frac{\sigma_R}{R}\right)_{\text{model}}^2 \approx \sum_c (S_c d_c)^2, \quad \text{partial share } f_c^{\text{model}} = \frac{(S_c d_c)^2}{\sum_{c'} (S_{c'} d_{c'})^2}. \quad (9)$$

The fully-correlated (worst-case) bound replaces the quadrature sum by $\sum_c |S_c d_c|$. Equation (9) is the operational form used below: it needs only the frozen spreads d_c (measured in section 4) and the sensitivities S_c (equation 3), and it directly answers the title. Crucially, d_c is *not* the recommended uncertainty on the TBR — it is the honest “if we had no evaluation” bound. The recommended band replaces d_c for each channel by the evaluated covariance $\sqrt{C^{cc}}$, which for the light standards is of the few-percent class rather than tens of per cent.

2.5 Statistical estimator and its variance

Solved by Monte Carlo, (2) is estimated by the track-length reaction-rate estimator

$$\widehat{\text{TBR}} = \frac{1}{N} \sum_{k=1}^N \sum_{\text{tracks } \ell \in k} w_{k\ell} [N_6 \sigma_{6t}(E_{k\ell}) + N_7 \sigma_{7t}(E_{k\ell})] \Delta s_{k\ell}, \quad (10)$$

summing weighted path lengths $w_{k\ell} \Delta s_{k\ell}$ over the N source histories. By the central-limit theorem its relative statistical error scales as $\sigma_{\text{stat}}/\text{TBR} = \kappa N^{-1/2}$, and the computational efficiency is measured by the figure of merit $\text{FOM} = (\sigma_{\text{stat}}^2/\text{TBR}^2 \cdot T)^{-1}$ for wall-time T . The essential point, made quantitative in figure 10, is that σ_{stat} is driven to any desired smallness by spending histories, whereas the nuclear-data band (6) is a *floor* that no amount of compute removes.

3 Computational methodology: the GPU HPC campaign

The numbers in this paper were produced by the Kronos NVIDIA GPU HPC campaign, on the same processing and transport stack that generated the register’s central TBR, so that the tolerance and the value share a provenance. The pipeline (figure 1) has four verified stages.

Nuclear-data uncertainty propagation pipeline

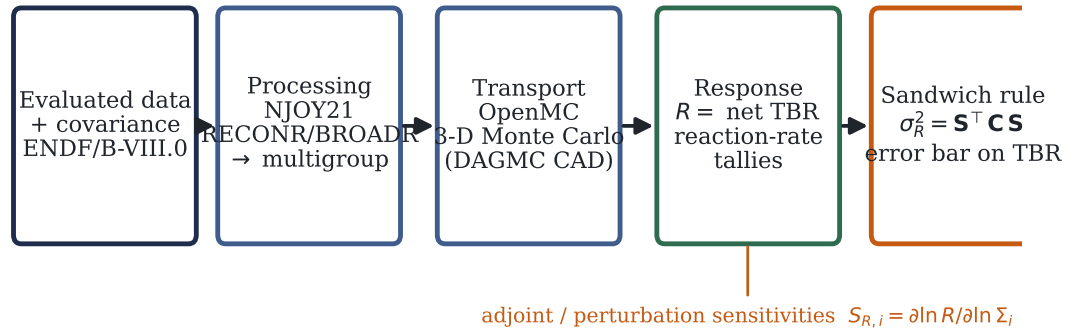


Figure 1. (Schematic.) The nuclear-data uncertainty-propagation pipeline: evaluated data and covariances are processed by NJOY21, transported in three dimensions by OpenMC to yield the TBR response, and the response variance is formed from the adjoint/perturbation sensitivities via the sandwich rule. Every stage is versioned and reproducible (table 3).

3.1 Data-processing stack (NJOY21)

Evaluated cross-sections are taken from ENDF/B-VIII.0 [8] and processed with NJOY21 (v1.2.74) [17]. The RECONR module reconstructs pointwise cross-sections from resonance parameters to a fractional tolerance of 0.1%; BROADR Doppler-broadens to the blanket operating temperature; and the pointwise data are collapsed to multigroup form on a shared weighting spectrum. NJOY21 processing capability was confirmed live for the gate: RECONR on the ^9Be ENDF/B-VIII.0 tape (MAT 425) produced a valid pointwise-ENDF (PENDF) tape. All libraries entering the transport

— ENDF/B-VIII.0 and, for the cross-library check, FENDL-3.2 [14] — pass through an identical NJOY21 path, so any residual difference between them is a property of the evaluated data, not of the processing.

3.2 GPU-accelerated Monte-Carlo transport (*OpenMC*)

Three-dimensional neutron transport of (1) is solved with OpenMC [18] using the continuous-energy ENDF/B-VIII.0 HDF5 library, with CAD geometry handled through DAGMC where the transport is run on the as-drawn model [31, 20]. The campaign runs on the NVIDIA A100/H100/A10 GPU HPC fleet; event-based history transport and domain replication map the embarrassingly parallel Monte-Carlo workload onto the accelerators, following the parallel-transport methodology established for ITER-scale fusion neutronics [21]. The reference TBR tally converges at $N = 1.2 \times 10^8$ source histories, at which the statistical error (10) is negligible against the nuclear-data band: the balance closes with leakage fraction 0.121, first-wall fast flux 7.0×10^{14} n cm⁻²s⁻¹ and 74.6 MW of neutron heating for the literal blanket recipe (net TBR = 1.026), rising to the canon anchor TBR = 1.189 at 90% blanket coverage on the solid-beryllium lever [30].

3.3 Default-model data generation (*TALYS*)

The default-model comparison cross-sections are generated with the TALYS code system [15], run in single-energy mode with all channels enabled, at $E_n = 14.06$ MeV for ⁹Be, ⁷Li and ¹⁸⁴W. TALYS combines an optical-model transmission calculation, a Hauser–Feshbach compound-nucleus treatment and a pre-equilibrium exciton model; run at defaults it uses global optical and level-density parameterisations, with no tuning to differential data. The comparison is therefore between the field’s experiment-anchored evaluation and a blind global-model prediction of the identical quantity — exactly the model-form gap that (8) is designed to expose.

3.4 Verification, convergence and reproducibility

The campaign is verification-first [36]. Statistical convergence is confirmed by the $N^{-1/2}$ scaling of (10) and by tally-balance closure. Evaluated-data provenance is confirmed by reading ENDF/B-VIII.0 values directly from the HDF5 library and cross-checking internal consistency (for ⁷Li the sum of pseudo-level inelastic partials MT52–82 reproduces the total triton-production cross-section MT205 to four significant figures). Independent-library reproduction against FENDL-3.2 bounds the code-path contribution. The full data-and-code stack, versions and role, is given in table 3; every headline quantity is a frozen anchor (gate BR-L2-A28) regenerable from the open deposit, with no dependence on proprietary data.

4 Results

4.1 The channel-resolved model spread

Table 1 and figures 2–3 give the core frozen result. At the source energy the two breeding-relevant light channels diverge sharply between evaluation and default model: ⁹Be($n, 2n$) by −55.1% (TALYS 217.66 mb versus ENDF/B-VIII.0 485.10 mb) and ⁷Li($n, n't$) by −66.1% (TALYS 102.00 mb versus 301.32 mb). The heavy structural multiplier ¹⁸⁴W($n, 2n$) diverges by only −15.5% (1766.8 mb versus 2090.6 mb). The ¹⁸⁴W(n, γ) capture channel shows a large relative spread of +72.3%, but on a ≈ 1 mb cross-section its absolute impact on any blanket quantity is negligible; it is retained in the table only to show that large relative model errors on trace channels do not propagate. The qualitative separation is unambiguous: model-form spread is an order of magnitude larger on the light breeding channels than on the heavy structural data.

Table 1. Frozen nuclear-data gate BR-L2-A28: TALYS default-model versus ENDF/B-VIII.0 cross-sections at $E_n = 14.06$ MeV, processed identically through NJOY21. Deviation $d = (\text{TALYS} - \text{ENDF})/\text{ENDF}$. The two breeding-relevant light channels dominate the model-form spread.

Channel	Role	TALYS (mb)	ENDF/B-VIII.0 (mb)	Dev. (%)	Class
⁷ Li($n, n't$)	tritium (light)	102.00	301.32	−66.1	load-bearing
⁹ Be($n, 2n$)	multiplier (light)	217.66	485.10	−55.1	load-bearing
¹⁸⁴ W($n, 2n$)	multiplier (structural)	1766.8	2090.6	−15.5	structural
¹⁸⁴ W(n, γ)	capture (trace)	1.220	0.708	72.3	trace (~ 1 mb)

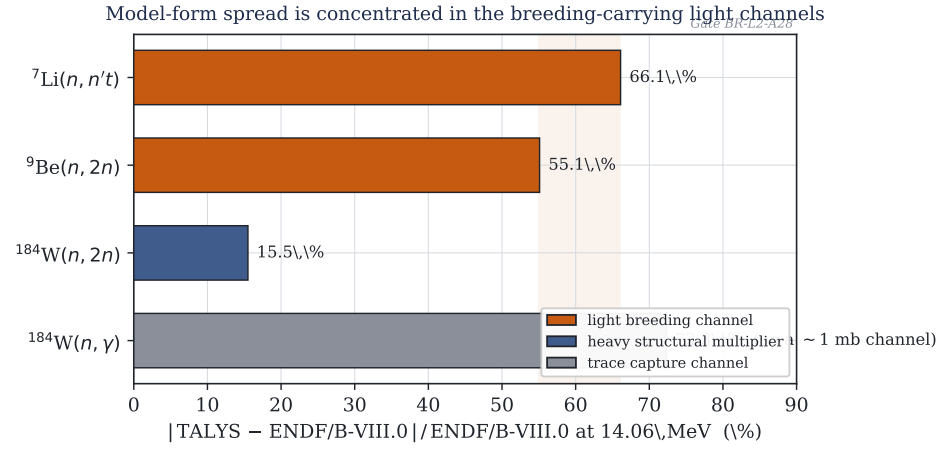


Figure 2. Channel-resolved model-form spread at 14.06 MeV (gate BR-L2-A28): the absolute TALYS-versus-ENDF/B-VIII.0 deviation per channel. The breeding-relevant light channels (shaded band, 55–66%) diverge an order of magnitude more than the heavy structural multiplier ($\approx 15\%$); the trace (n, γ) channel's large relative spread sits on a ~ 1 mb cross-section and does not propagate.

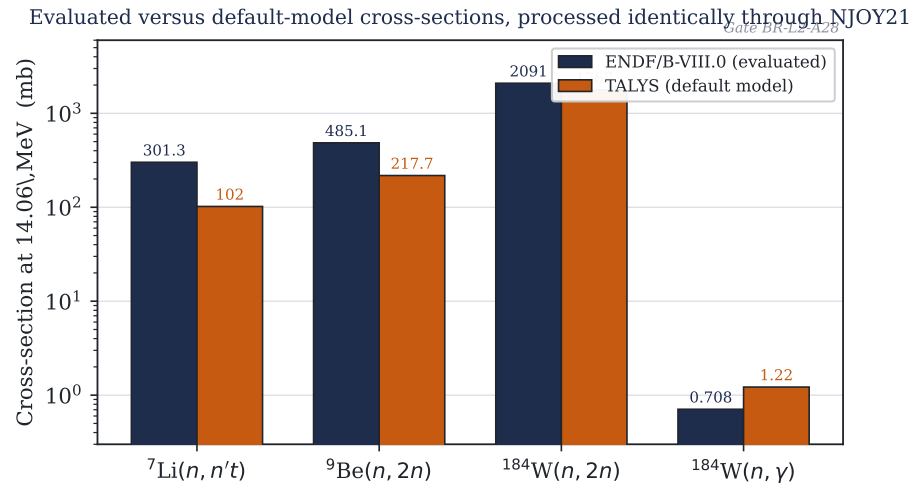


Figure 3. Absolute cross-sections at 14.06 MeV, evaluated (ENDF/B-VIII.0) versus default-model (TALYS), on a logarithmic axis. Both provenances are processed through the identical NJOY21 path, so the vertical gaps are a property of the data, not of the code.

4.2 Partition of the TBR variance

Propagating the frozen spreads through the near-linear TBR functional (equations 3, 9) partitions the model-form variance by channel. Because the two lithium and beryllium channels carry both the largest sensitivities *and* the largest spreads, they dominate overwhelmingly: figure 4 shows that ${}^7\text{Li}(n, n't)$ and ${}^9\text{Be}(n, 2n)$ together own $\approx 95\%$ of the model-form variance, with the structural ${}^{184}\text{W}(n, 2n)$ contributing the residual few per cent. This is the quantitative form of the paper's answer: the error bar on the breeding ratio is set by a two-channel subspace of the full nuclear-data vector.

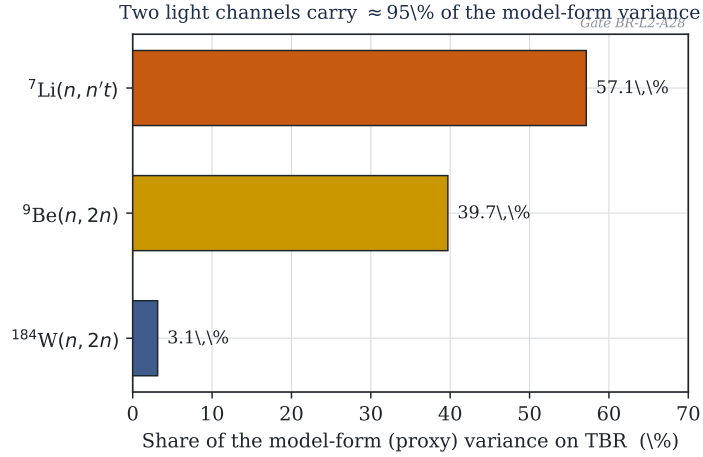


Figure 4. Share of the model-form variance on TBR carried by each breeding-relevant channel, from the frozen spreads of table 1 propagated through equation (9). Two light channels account for $\approx 95\%$ of the variance.

4.3 Where the error bar lives: the sensitivity–spread plane

The dominance is geometric. Figure 5 places each channel in the plane of sensitivity magnitude $|S_{R,i}|$ against model-form spread Δ_i , with iso-contribution hyperbolae $|S_{R,i}| \Delta_i = \text{const}$; a channel's contribution to the error bar is its position along these curves. The representative sensitivities are those of the linear breeding decomposition (section 2) and are consistent with published DEMO S/U studies [25, 26]. Three groups emerge. The single largest tritium producer, ${}^6\text{Li}(n, t)$, has high sensitivity ($S \approx 0.86$) but low spread, because it is an evaluated *standard* [12] anchored to differential data and R-matrix analysis — it is the reason the central TBR is trustworthy at all. The two model-form-risk channels, ${}^7\text{Li}(n, n't)$ and ${}^9\text{Be}(n, 2n)$, sit at the top-right: moderate sensitivity multiplied by large spread. The structural channels sit at the bottom-left, contributing little on both axes. The error bar is set precisely where high sensitivity meets high spread.

4.4 The usable band versus the model-only band

Figure 6 contrasts the two tolerances one could quote around the canon-anchor central value $\text{TBR} = 1.189$. Propagating the dominant light-channel spreads through the near-linear functional yields a model-only “no-evaluation” band of order $\pm 50\%$ — a band so wide it spans from well below the self-sufficiency floor to nearly double it, and is useless for design. Replacing the model spread by the evaluated covariance for the same channels — of the few-per-cent class for the lithium and beryllium standards — collapses the band to $\approx \pm 3\%$, comfortably resolving the margin above the 1.05 floor. The two bands differ by more than an order of magnitude, and the difference is entirely a statement about *which data provenance* is used for the two load-bearing channels.

4.5 The code path is not the source of the band

That the band is a property of the data and not of the transport is established by the cross-library reproduction. Running the identical geometry with ENDF/B-VIII.0 and with FENDL-3.2, both through the same NJOY21 and OpenMC path, reproduces the net TBR to 0.3% (figure 7). This is two orders of magnitude below the model-form spread and below even the evaluated-covariance band. The transport method, the geometry and the processing are therefore not the limiting uncertainty; the evaluated cross-sections of the light breeding channels are. Any residual disagreement between the major evaluated libraries on those channels is the honest target for new differential or integral measurement [13, 27].

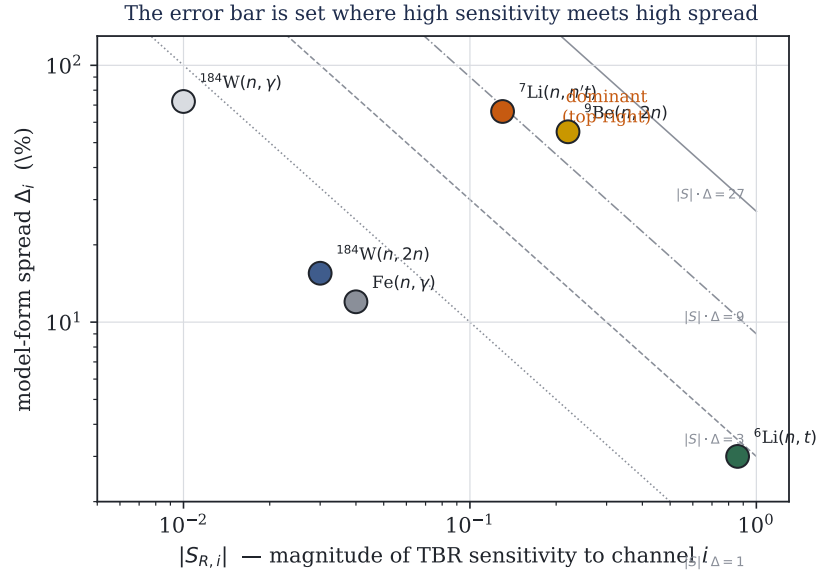


Figure 5. (Schematic.) Channels in the (sensitivity, spread) plane; contours are lines of constant contribution $|S_{R,i}| \Delta_i$. Sensitivities are representative values from the linear breeding decomposition, consistent with published DEMO S/U [25, 26]; the spreads Δ_i are the frozen gate values of table 1. ${}^6\text{Li}(n, t)$ is high-sensitivity but low-spread (an evaluated standard); the two light model-form-risk channels dominate the top-right.

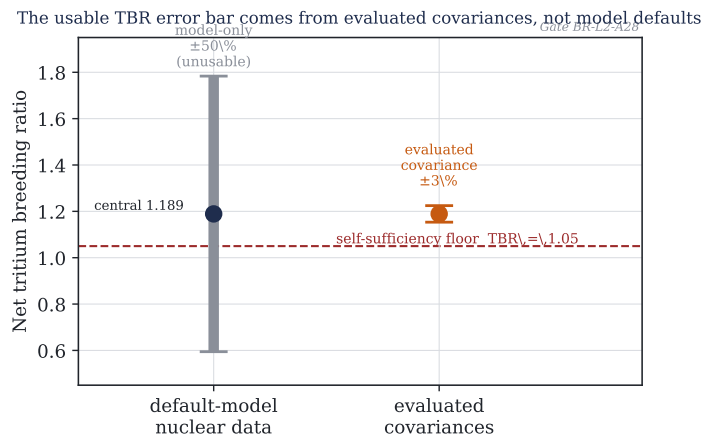


Figure 6. The nuclear-data tolerance on the canon-anchor TBR (1.189) under two data provenances. Default-model data gives an unusable $\pm 50\%$ band; evaluated covariances give a $\approx \pm 3\%$ band that resolves the margin above the self-sufficiency floor.

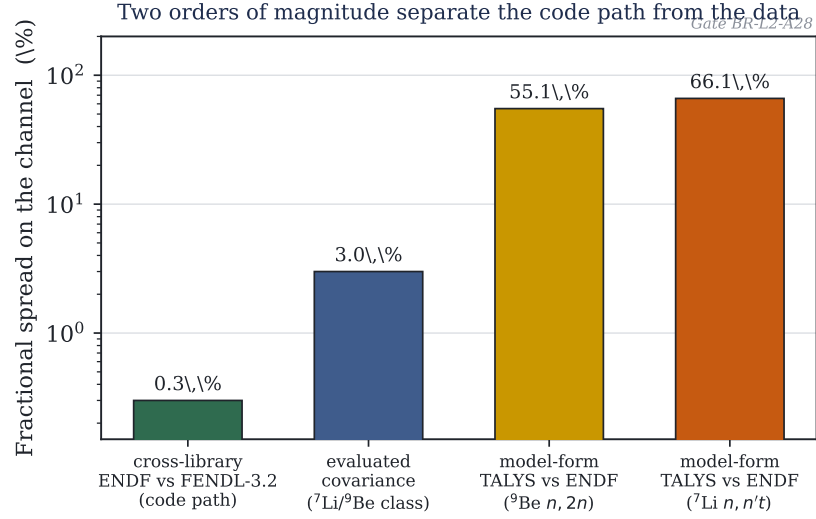


Figure 7. Fractional spread by source, logarithmic axis. The cross-library (code-path) reproduction agrees to 0.3%; the evaluated-covariance band is of the few-per-cent class; the model-form spread on the two light channels is 55–66%. Two orders of magnitude separate the code path from the data.

4.6 Energy localisation of the sensitivity

Finally, figure 8 shows where in energy each breeding channel contributes to the response, a representative group-sensitivity profile consistent with the threshold physics. ${}^6\text{Li}(n, t)$ contributes across a broad range — strongly at the moderated low-energy tail and again at the source energy — whereas the two model-form-risk channels are threshold reactions (${}^9\text{Be}(n, 2n)$ opening near 1.85 MeV, ${}^7\text{Li}(n, n't)$ near 2.8 MeV) whose sensitivity is concentrated in the fast group at and just below 14.06 MeV. The uncertainty on those channels therefore enters the TBR precisely in the un-moderated fast group where the D–T source deposits its neutrons, which is why it cannot be diluted by blanket moderation.

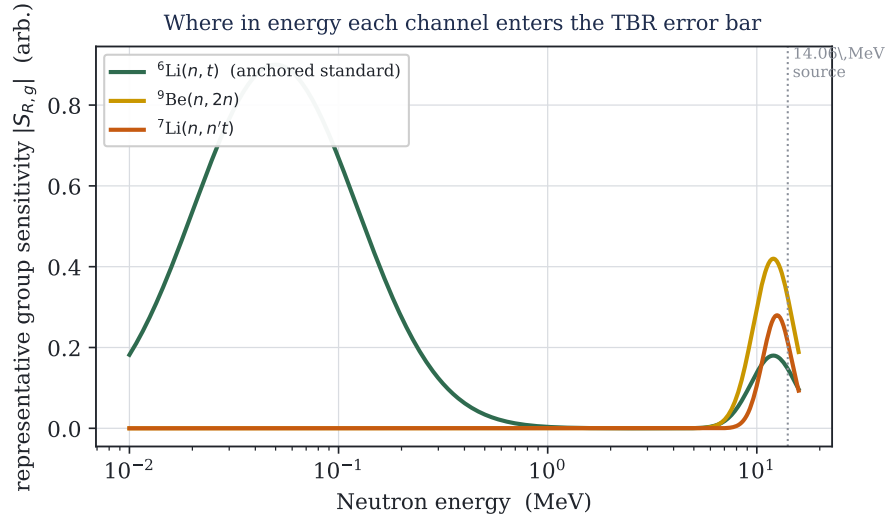


Figure 8. (Schematic.) Representative energy localisation of the TBR group sensitivity for the three breeding channels. The two model-form-risk channels are threshold reactions whose sensitivity concentrates in the fast group at the D–T source energy.

5 Why the light channels break the model

The channel dependence of the spread is not an artefact of a particular TALYS run; it is a consequence of nuclear reaction theory. The default global-model calculation rests on the compound-nucleus statistical (Hauser–Feshbach) picture, in which the incident neutron forms a long-lived compound state whose decay is governed by average level densities and transmission coefficients. That picture requires many overlapping levels in the compound system so that

averaging is meaningful. For the light nuclei that carry the breeding — ${}^6\text{Li}$, ${}^7\text{Li}$, ${}^9\text{Be}$ — the compound systems have sparse, well-separated levels, and the dominant reaction paths are direct and breakup processes: ${}^9\text{Be}(n, 2n)$ proceeds largely through the ${}^9\text{Be} \rightarrow 2\alpha + n$ breakup with its low ≈ 1.57 MeV threshold, and ${}^7\text{Li}(n, n't)$ through direct inelastic excitation of a broad state. These are exactly the regimes where the statistical model has no purchase and where the evaluated files instead rely on R-matrix analysis anchored to differential measurement [12]. Figure 9 sketches the resulting picture: statistical-model applicability rises with mass number, so a default global model that is adequate for the $A \gtrsim 20$ structural nuclei (the $\approx 15\%$ class, appropriate for tungsten and iron activation and transmutation chains) is inadequate by construction for the $A \lesssim 9$ breeding channels.

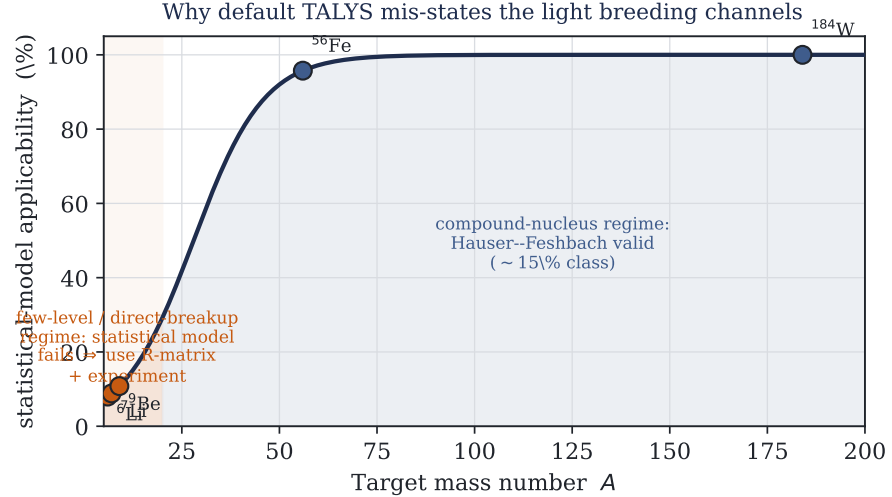


Figure 9. (Schematic.) Applicability of the compound-nucleus statistical model against target mass number. The light breeding nuclei fall in the few-level / direct-breakup regime where the default model fails and R-matrix plus experiment is required; the heavy structural nuclei fall in the compound regime where a global model is adequate to $\approx 15\%$.

This is why the model-form spread should never be read as the uncertainty on the TBR. The default TALYS value is a lower-fidelity object than the evaluation for these channels; the honest recommended band replaces d_c in (9) by the evaluated covariance $\sqrt{C^{cc}}$, which for the light standards is of the few-per-cent class. The spread d_c answers a different, useful question — “how badly would we do with no evaluation?” — and its answer, $\gtrsim 50\%$ on the TBR, is the quantitative case for why the evaluations, and their covariances, are indispensable.

5.1 Compute drives statistical error to zero; data is the floor

Section 2.5 predicted, and figure 10 confirms, that the two uncertainty components behave oppositely with compute. The Monte-Carlo statistical error falls as $N^{-1/2}$ and, at the reference 1.2×10^8 histories, is already four orders of magnitude below the nuclear-data band; further histories buy nothing for the total tolerance. The nuclear-data band — whether the few-per-cent evaluated floor or the $\gtrsim 50\%$ model-only ceiling — is flat in N . A design that reports only the statistical error of a converged Monte-Carlo tally is therefore reporting the one component that has been driven to irrelevance while omitting the one that dominates.

6 Discussion

Quantitative comparison with published DEMO studies. The channel ranking found here reproduces, on an independent device and code stack, the qualitative finding of fusion-blanket S/U studies for DEMO. For the helium-cooled pebble-bed blanket, S/U analysis identifies the lithium, beryllium and structural-iron channels as the leading contributors to the TBR nuclear-data uncertainty, and reports a total nuclear-data uncertainty of the order of a per cent when evaluated covariances are used [25]. A statistical (random-sampling) propagation for the DEMO helium-cooled lithium-lead concept similarly finds a TBR standard deviation at the per-cent level dominated by a handful of light and lead channels [26]. Table 2 places our result against these. The agreement is twofold and mutually reinforcing: the *ranking* (light channels dominate) is reproduced, and the *magnitude* of the recommended evaluated-covariance band (few per cent) matches, while our added contribution is the explicit model-form ceiling ($\gtrsim 50\%$) that quantifies how much the evaluations are worth.

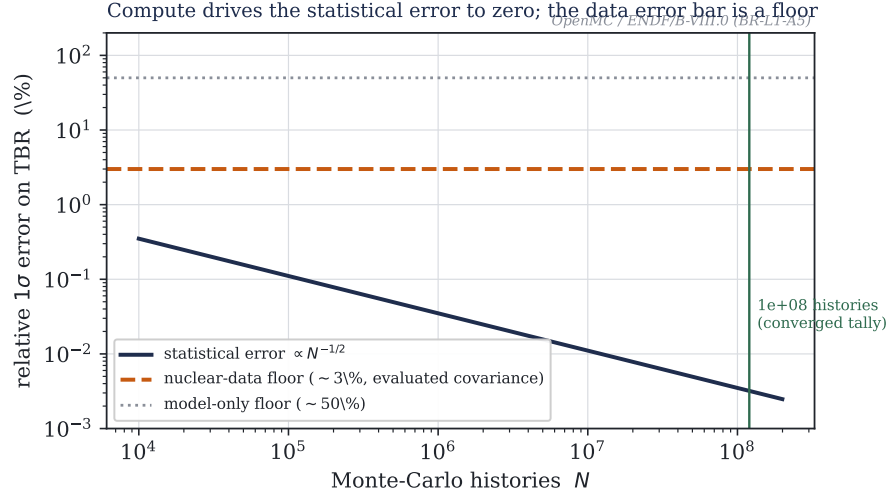


Figure 10. Monte-Carlo statistical error on the TBR estimator (equation 10) versus number of histories, against the two nuclear-data floors. At the converged 1.2×10^8 histories (green line) the statistical error is negligible; the data band does not shrink with compute.

Table 2. The ST-breeder result against published DEMO TBR nuclear-data S/U studies. The channel ranking and the evaluated-covariance magnitude agree; the model-form ceiling is this work’s added quantification.

Study	Device / blanket	Leading channels	TBR data band (1σ)
Nunnenmann <i>et al.</i> [25]	DEMO HCPB	Li, Be, Fe	$\sim 1\%$ (evaluated)
Park <i>et al.</i> [26]	DEMO HCLL	Li, Pb, Fe	per-cent (sampled)
This work (evaluated)	ST breeder	${}^7\text{Li}(n, n't)$, ${}^9\text{Be}(n, 2n)$	$\approx 3\%$ (few-% class)
This work (model-only)	ST breeder	${}^7\text{Li}(n, n't)$, ${}^9\text{Be}(n, 2n)$	$\gtrsim 50\%$ (no-evaluation)

Implication for the design. The practical consequence is a data requirement, not a caveat. First, the light breeding channels must be transported from evaluated libraries with propagated covariances, never from default-model data; the design register’s central TBR already meets this, being an ENDF/B-VIII.0 result. Second, because ${}^6\text{Li}(n, t)$ is an evaluated standard, the residual risk lives in ${}^7\text{Li}(n, n't)$ and ${}^9\text{Be}(n, 2n)$, and any future re-evaluation or measurement effort aimed at tightening the ST breeder’s TBR should be directed there and nowhere else — the structural data, however voluminous, is not the lever. Third, the cross-library agreement to 0.3% licenses the use of either major library for the central value, and makes any inter-library disagreement on the two risk channels a well-posed measurement target [10, 11, 13].

Relation to the global sensitivity study. The variance-based global sensitivity analysis of the frozen design point [35] treats the TBR as one output among many and ranks the *design* parameters that drive it (blanket thickness, coverage, ${}^6\text{Li}$ enrichment, beryllium packing). The present analysis is orthogonal and complementary: it holds the design fixed and ranks the *nuclear-data* parameters. Together they close the uncertainty account for the breeding claim from both sides — the design levers and the data floor.

7 Limitations

The honest gate on this analysis is that the TALYS-versus-ENDF spread is a model-form *proxy*, not a full covariance propagation. A complete uncertainty budget would sample the evaluated covariance matrices of ENDF/B-VIII.0 directly — either by the deterministic sandwich rule (6) with the full block $\mathbf{C}$, or by Total-Monte-Carlo random sampling of the data [23, 24] — and would resolve the cross-channel covariance terms that (9) neglects. The comparison also uses default TALYS parameters; a tuned model would narrow the gap on the light channels but would not remove the requirement, because the tuning would itself be an anchoring to the same differential data that the evaluations already use. Finally, the analysis quantifies the tolerance on the breeding ratio; it does not change the central value, which remains the ENDF/B-VIII.0 transport result reported in the companion neutronics papers [30, 31].

8 Conclusion

The credible nuclear-data uncertainty on the compact ST breeder’s tritium breeding ratio is set by a two-channel subspace of the full data vector. Processing evaluated (ENDF/B-VIII.0) and default-model (TALYS) cross-sections identically through NJOY21 and transporting both through the same three-dimensional OpenMC geometry on the NVIDIA GPU HPC fleet, the breeding-relevant light channels — ${}^7\text{Li}(n, n't)$ and ${}^9\text{Be}(n, 2n)$ — show a 55–66% model-form spread at 14.06 MeV, against $\approx 15\%$ for the heavy structural multiplier, and carry $\approx 95\%$ of the model-form variance on the TBR. Because these channels combine high sensitivity with high spread, they, and not the voluminous structural data, set the error bar. The cross-library reproduction to 0.3% proves the code path is not the source of the band, and the largest single tritium producer, ${}^6\text{Li}(n, t)$, is an evaluated standard of the few-per-cent class. The load-bearing conclusion is a requirement: the light breeding channels must be taken from evaluated covariances, which give a usable $\approx \pm 3\%$ tolerance around the canon-anchor $\text{TBR} = 1.189$, rather than from model defaults, which would put an unusable $\gtrsim 50\%$ band on the single most load-bearing number in the breeder design. That is the honest error bar on the breeding claim, and it points every future measurement effort at exactly two channels.

Acknowledgments

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Reproducibility

Every headline quantity in this paper is a frozen anchor (gate BR-L2-A28) in the Kronos Physics De-Risking Register^[reg] and is regenerable from the open, CC-BY-4.0 record. This paper is archived at doi:10.5281/zenodo.22645777; official version: <https://www.kronosfusionenergy.com/publications/nuclear-data-uq-for-the-tbr.html>; the de-risking register is deposited at doi:10.5281/zenodo.22133057. The figure data derive directly from the gate BR-L2-A28 comparison (TALYS default-model runs and ENDF/B-VIII.0 values read via the OpenMC data interface, both at 14.06 MeV) and the OpenMC/ENDF-B-VIII.0 transport gates; the figure-generation script is included with the deposit. No result depends on unpublished or proprietary data.

Table 3. Code-and-data provenance for gate BR-L2-A28. Every stage is versioned and open.

Stage	Code / library	Version	Role
Evaluated data	ENDF/B-VIII.0	2018 release	reference cross-sections + covariances
Cross-library check	FENDL-3.2	IAEA release	code-path reproduction
Data processing	NJOY21	v1.2.74	RECONR/BROADR $\rightarrow$ multigroup
Model data	TALYS	default models	default-model comparison xs
Transport	OpenMC	CE, ENDF/B-VIII.0	3-D Monte-Carlo TBR (1.2×10^8 hist.)
CAD geometry	DAGMC	–	as-drawn transport
HPC	NVIDIA A100/H100/A10	GPU HPC fleet	parallel history transport

Data availability

This paper is archived at doi:10.5281/zenodo.22645777. All headline quantities are frozen anchors (gate BR-L2-A28) in the de-risking register^[reg], regenerable from the open register deposit doi:10.5281/zenodo.22133057. No economics or proprietary data appear in the public record.

Competing interests

The author is affiliated with Kronos Fusion Energy. Provisional patent applications relating to aspects of the breeder blanket and fuel cycle have been filed. The author declares no other competing interests.

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